

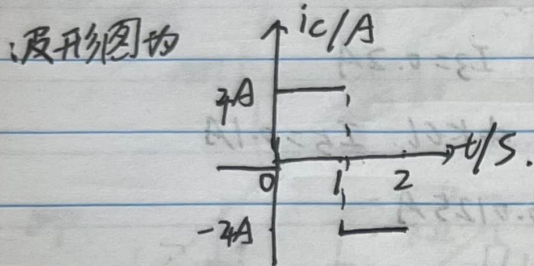
第4次作业

$t < 0 / t > 2$ 时 $p = 0$

1.2 (1) $i_C(t) = C \frac{du}{dt}$ $t < 0$ $t > 2$ 时 $i = 0A$

当 $t \in [0, 1]$ 时, $i_C(t) = 2A$

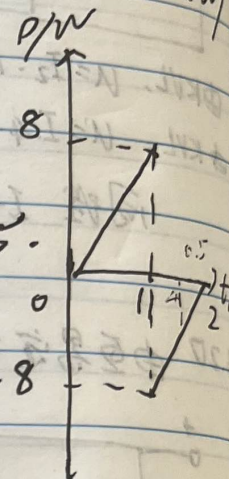
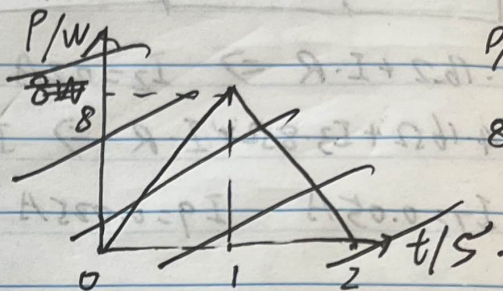
$t \in [1, 2]$ 时 $i_C(t) = -2A$



(2) 由 $p = u i$ 知

当 $t \in [0, 1]$ 时 $p = u_C(t) i_C(t) = 8t (W)$

$t \in [1, 2]$ 时 $p = u_C(t) i_C(t) = -16 + 8t (W)$



(3) 发出功率, $P = 24W$

储能 $Q = -\int_{1.5}^2 p dt = 1J$

1.3 (1) 由 $i_L(t) = L \frac{di}{dt}$

$\Rightarrow i_L(t) = \int \frac{1}{L} u_L(t) dt$

当 $t \in [0, 2]$ 时 $i_L(t) = \frac{5}{2}t$

当 $t \in [2, 3]$ 时 $i_L(t) = 5A$

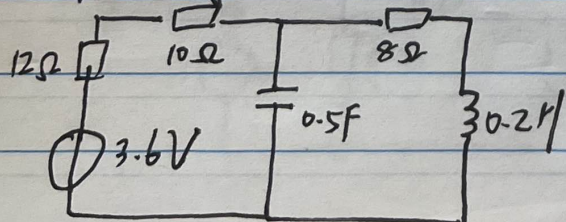
当 $t \in [3, 4]$ 时 $i_L(t) = -10t + \frac{5}{4}t^2 + \frac{95}{4}$

$t > 4$ 时 $i_L(t) = 3.75A$

(2) $W = \int_0^2 u i dt = \int_0^2 25t dt = -50J$

$\therefore Q = W = 50J$

1.5 稳定时, 电流源 $\rightarrow$ 电压源, 转化电路如下.



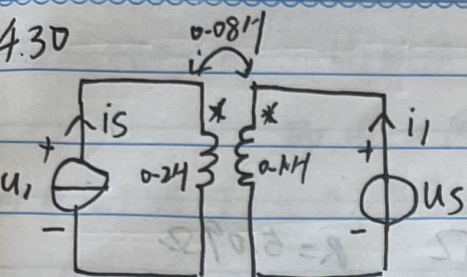
$\therefore I = \frac{U}{R_{\Sigma}} = 0.12A$

电阻上电压 $U = 8\Omega \cdot I = 0.96V$

$\therefore P_L = \frac{1}{2} I^2 = 1.44 \times 10^{-3} W$

$P_C = \frac{1}{2} C U^2 = 0.2304 J$

4.30



$$L_1 = 0.2H \quad L_2 = 0.1H \quad M = 0.08H$$

有 $u_1 = i_s$

$$u_1 = L_1 \frac{di_1}{dt} + M \frac{di_2}{dt}$$

$$u_2 = L_2 \frac{di_2}{dt} + M \frac{di_1}{dt}$$

代入有

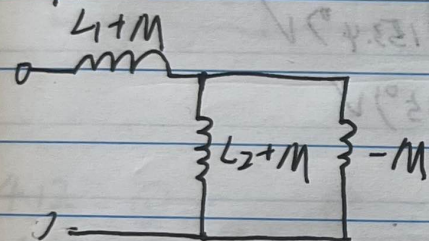
$$u_1 = 0.2 \cdot (1 - 6e^{-10t}) + 0.08 \frac{di_2}{dt}$$

$$10te^{-20t} = L_2 \frac{di_2}{dt} + 0.08 \cdot (1 - 6e^{-10t})$$

$$\Rightarrow u_1 = -1.2e^{-10t} + 8te^{-20t} + 0.384e^{-10t}$$

$$= 8te^{-20t} - 0.816e^{-10t} \quad \checkmark$$

4.31 等效电路图为



$$L_{eq} = L_1 + M + \frac{(L_2 + M) \cdot (-M)}{L_2 + M - M} = \frac{L_2 - M - M^2}{L_2}$$

$$= L_1 - \frac{M^2}{L_2}$$

4.32 $I_2 = \frac{U_2}{16\Omega}$

$$\therefore \frac{I_1}{I_2} = \frac{1}{n}$$

$$\Rightarrow I_1 = \frac{U_2}{n \cdot 16\Omega}$$

$$\therefore U_1 = 4\Omega \cdot I_1 + nU_2$$

$$\text{令 } U_1 = U_2 \text{ 时 } n = \frac{1}{2}$$